EPIDEMIOLOGICAL MACHINE LEARNING • POPULATION SCREENING TRIAGE

When Training Labels Change Who Gets Screened

Temporal evaluation of diagnosis-based versus biomarker-augmented labels for diabetes screening in NHANES

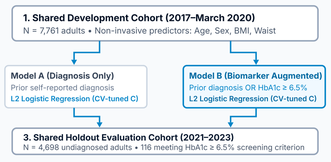
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01 THE CLINICAL PROBLEM

Population screening triage algorithms frequently train on self-reported prior diagnoses or electronic health record proxies. However, clinical diagnoses require regular clinical encounters, misclassifying unrecorded disease as healthy baselines among underserved individuals facing access barriers.

02 STUDY DESIGN & TEMPORAL PROTOCOL

We evaluate whether augmenting self-reported diagnosis with objective venous HbA1c (glycated hemoglobin ≥ 6.5%) alters triage priorities in CDC NHANES (National Health and Nutrition Examination Survey), holding predictors and capacity constant.



03 METHODS & IMPLEMENTATION NOTE

Primary models are **L2-regularized logistic regressions** tuned via 5-fold cross-validation on median-imputed, standard-scaled numeric features (age, BMI, waist circumference) and binary one-hot encoded sex. Poverty Income Ratio (PIR) and insurance were strictly excluded from model inputs and reserved solely for equity audits. Data pipeline ingested 10 XPT files across two survey periods using an embedded DuckDB lakehouse; Pandera contracts validated biological ranges (HbA1c: 2.0%–25.0%) and confirmed 0 pregnant participants.

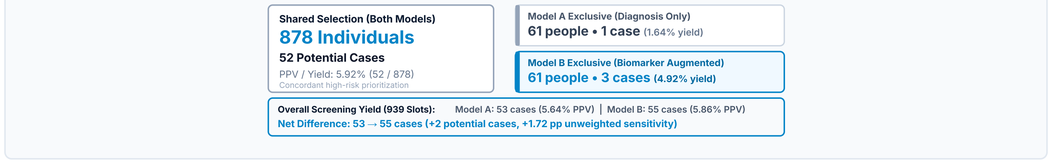
KEY FINDING

Changing training labels altered screening priorities and identified two additional potential cases at the same screening capacity.

04 PAIRED-SELECTION QUEUE ANALYSIS (20% SCREENING CAPACITY)

Fixed Capacity K = 939 slots / 4,698 adults

At 20% screening capacity (939 test slots), both models prioritize 878 shared individuals. Priorities diverge for 122 individuals at the margin:



Schematic: elements not to scale. Positive Predictive Value (PPV) is the percentage of screened individuals who meet the HbA1c criterion. The 3.0× yield difference (4.92% vs 1.64%) applies strictly to the 61-person boundary swap, not to overall model precision (5.86% vs 5.64% unweighted). Overall case detection increased from 53 to 55 (+2 potential cases).

BENCHMARK PERFORMANCE & COMPLEX SURVEY VARIANCE (20% CAPACITY)

Screening Model	Slots (K)	Unweighted Cases	Unweighted Sens	Survey-Weighted Sens (95% CI)	Survey-Weighted PPV (95% CI)
Model A (Diagnosis Only)	939	53 / 116	45.69%	40.08% [28.9%, 51.2%]	5.79% [3.6%, 8.0%]
Model B (Biomarker Augmented)	**939**	**55 / 116**	**47.41%**	**42.47% [28.4%, 56.5%]**	**6.02% [3.4%, 8.6%]**
Net Change (Δ)	0	**+2 Cases**	**+1.72 pp**	**+2.38 pp** (overlapping CIs)	**+0.22 pp** (overlapping CIs)

Survey inference note: Weighted estimates use Taylor Series Linearization (WTPH2YR weights, df = 15). The observed point-estimate difference is small (+2 cases, +2.38 pp weighted sensitivity), and formal paired confidence intervals accounting for complex survey clustering have not been evaluated.

05 HEALTH EQUITY SUBGROUP AUDIT

Evaluating detection across insurance status at 20% budget:

Subgroup	Total Cases	Model A Detection	Model B Detection
Insured Adults	102 cases	51 / 102 (50.0%)	52 / 102 (51.0%)
Uninsured Adults	**13 cases**	**2 / 13 (15.4%)**	**3 / 13 (23.1%)**

Small denominator note: The uninsured cohort contains only **13 latent cases**; detection changed by +1 case (2/13 to 3/13). While directionally consistent with hypothesis, small cell counts preclude claims of disparity elimination or causal improvement.

06 MULTI-ARCHITECTURE ROBUSTNESS

Triage re-evaluated under alternative tree architectures:

- **Random Forest:** Model A detects 60 cases; Model B detects 61 cases (+1 case, 77 swaps).
- **HistGradientBoosting:** Model A detects 59 cases; Model B detects 59 cases (0 net gain, 132 swaps).

Confirms priority shifts occur across architectures, while overall yield gains remain modest (+0 to +2 cases).

07 LIMITATIONS & SCOPE

- **Surveillance vs Diagnosis:** Single venous HbA1c ≥ 6.5% represents an elevated screening criterion, not an autonomous clinical diagnosis.
- **Inference Bounds:** Separate survey-weighted confidence intervals overlap; paired variance estimation is required for confirmatory testing.
- **Operational Scope:** Assumes 100% screening adherence among prioritized adults and uniform laboratory costs.

08 CONCLUSION & NEXT STEPS

Augmenting administrative diagnosis labels with objective biomarker ground truth modestly refines borderline triage (+2 cases, 53 → 55) without expanding predictor requirements. Next steps require evaluating formal paired complex survey variance and conducting prospective validation in diverse clinical care settings.

References (APA 7th): American Diabetes Association. (2024). *Diabetes Care*, 47(Suppl. 1), S20–S42 • CDC NCHS. (2024). *NHANES 2017–2023*. U.S. DHHS • Obermeyer et al. (2019). *Science*, 366(6464), 447–453

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